J C Bose University of Science and Technology, YMCA Faridabad, (Deptt of Mech. Engg.)

Subject -- AT　　　　Sessional Test-I, (2024)　　　　MM- 15

Class IVth SM Mech. Engg.　　(M41&M42)　　　　Time- 90 Min

Attempt any three questions. All questions carry equal marks. Steam table may be used.

1. What is stoichiometry? Calculate the static draught produced by a chimney of 40 m height for maximum rate of discharge of hot gas through it. The atmospheric air temperature may be taken as 30°C. (5) (CO=1)

2. Explain the Artificial draught, their types and applications. Differentiate between Lamont boiler and Velox boilers in their construction and uses. (5) (CO=1).

3. Explain the boiler mountings and accessories. The following data refer to a boiler trial. The analysis of coal gives C - 81 %, H_2=4.5%, 02- 8%, and remainder incombustible. The analysis of dry flue gas gives CO_2-8.3%, CO-1.4%, N_2-80.3% and 0_2= 10%. Calculate (a) the mass of air supplied per kg of fuel (i.e. Air-fuel ratio) (5) (CO) = 2

4. Explain the Rankine vapour power cycles. In a Rankine cycle steam enters the turbine at 140 bar and 550°C. If the exhaust pressure is 0.06 bar and all processes are reversible, (a) Find the cycle thermal efficiency, the work ratio and the steam rate (b) If the efficiency of turbine is 80% and the efficiency of pump is 50%, than what is thermal efficiency? (5) (CO=2)